

SQL Practical Practice Questions with Database Schema

Use the following tables to solve all practice questions.

Employees Table

- emp_id (INT) PRIMARY KEY
- name (VARCHAR(100))
- email (VARCHAR(100))
- salary (INT)
- department_id (INT)
- joining_date (DATE)
- city (VARCHAR(50))
- manager_id (INT)

Departments Table

- department_id (INT) PRIMARY KEY
- department_name (VARCHAR(100))

Customers Table

- customer_id (INT) PRIMARY KEY
- name (VARCHAR(100))
- email (VARCHAR(100))
- city (VARCHAR(50))

Orders Table

- order_id (INT) PRIMARY KEY
- customer_id (INT)
- amount (DECIMAL(10,2))
- order_date (DATE)
- status (VARCHAR(30))

Beginner Level

- 1 Display all employee details.
- 2 Display only name, salary, and city.
- 3 Find employees whose salary is greater than 50000.
- 4 Find employees belonging to Mumbai.
- 5 Find employees whose salary is between 50000 and 80000.
- 6 Display employees whose names start with A.
- 7 Display employees whose names end with a.

- 8 Display employees with salaries 60000, 70000, or 90000 without OR.
- 9 Sort employees by salary in ascending and descending order.
- 10 Display the first five employees.

Aggregate Functions

- 1 Find the maximum, minimum, average, and total salary.
- 2 Count total employees.
- 3 Find the total number of employees in each department.
- 4 Find departments having more than five employees.
- 5 Find the average salary of every department.

GROUP BY and HAVING

- 1 Find departments where average salary is greater than 80000.
- 2 Find cities having more than ten employees.
- 3 Display departments whose maximum salary exceeds 100000.

JOIN Questions

- 1 Display employee name, department name, and salary using JOIN.
- 2 Find employees who do not belong to any department.
- 3 Display department name and total employees.
- 4 Display all departments even if they have no employees.

Customers and Orders

- 1 Display customer name, order amount, and order date.
- 2 Find customers who never placed any orders.
- 3 Find customers having more than five orders.
- 4 Find the highest order amount.
- 5 Find total sales.
- 6 Display the top five customers having the highest purchases.

Intermediate Level

- 1 Find the second highest salary.
- 2 Find the third highest salary.
- 3 Find employees earning more than the average salary.
- 4 Find duplicate email addresses.
- 5 Delete duplicate employee records while keeping one record.
- 6 Find employees who joined in 2026.
- 7 Find employees with NULL manager_id values.

- 8 Find managers having more than five employees.

Advanced Level

- 1 Find the highest salary from every department.
- 2 Find the second highest salary from every department.
- 3 Display the top three salaries without using LIMIT.
- 4 Find customers who purchased more than 100000 in total.
- 5 Generate month-wise sales reports.
- 6 Generate daily sales reports.
- 7 Find the top-selling month.
- 8 Display employees whose salary is greater than their manager's salary.
- 9 Find employees having the same salary.
- 10 Generate department-wise highest salary, lowest salary, average salary, total employees, and total salary reports.

Bonus Challenge Questions

- 1 Build a sales dashboard query.
- 2 Find inactive customers (no orders in the last six months).
- 3 Find the employee with the longest tenure.
- 4 Find departments where no employee earns less than 50000.
- 5 Find the top 10% highest-paid employees.
- 6 Calculate running totals of sales using window functions.
- 7 Rank employees department-wise based on salary.
- 8 Find the percentage contribution of each department to the company's total salary expense.